

IOT6002U Assessment Brief – 002

Research Output

Module Code	IOT6002U
Module Title	Research Informed Professional Practice
Module Organiser(s)	Jonathan Jackson & Alex Cline
Assessment	002 Research Output
Weighting (%)	40%
Deadline	Tuesday 19 th January 2027, 23:59
Duration / Length	Upwards of 30 hours effort

Assessment Outline

For this assessment, you will individually produce a research output in whichever of the following formats you prefer:

- Traditional academic paper** adhering to the style and citation guidelines for a journal or publisher of your choice (e.g. IEEE)
- Bespoke interactive website** (publicly accessible hyperlink)
- Wiki or online knowledgebase** e.g. GitHub Wiki, Notion (publicly accessible hyperlink)

For your final submission, you will be required to submit **two** documents:

- Your final research output** which will be one of the following:
 - your completed paper OR
 - a document containing all the text and references used in your online artefact, along with a link to the artefact (website or wiki)
- Your research process evidence** which will include:
 - Search strategy, keywords, databases and tools used
 - Filtered or sampled search results, inclusion / exclusion criteria, notes, and evidence of how you refined the search results over time
 - Repeatable research process documentation
 - 5 annotated pages of a peer-reviewed academic paper
 - See the “Research Process” section below for more details

Research Focus

You may wish to align your research to the research plan from your 001 Project Proposal submission for this module, but this is not mandatory. The benefit of doing so will mean you can deepen your knowledge and expertise around topics related to your final project. You will not be asked to complete a full literature review for your Final Project report (for EPA), so this reduces the risk of self-plagiarism (using the same work for two assessments).

If you wish to choose a different area of research, it must be aligned with your apprenticeship or professional context in some way. Example research topics might include the following, all of which can be linked to at least one KSB from your apprenticeship standard:

- An emerging software development tool (K5, S14)
- Agentic AI (S14)
- Digital Ethics (K19)
- Workplace Inclusivity (B8)
- Recent cyber security incidents (K11)

You should also specify at least one research question which your research will help answer e.g. “What are the potential impacts of GPT-6 on the cyber security sector in the next 3 years?” or “How might the potential ethical impacts of Agentic AI vary between Europe, Africa, and China based on cultural differences?”

Research Scope

You are **not** expected to carry out primary data collection, such as surveys, interviews, focus groups, or user testing, for this assessment. Your research should be based on existing sources.

You should draw on a minimum of **15 credible sources** in the final output, of which at least **5 should be peer-reviewed academic sources**. Other sources may include professional or industry reports, technical documentation, standards, policy documents, credible datasets, pre-prints, organisationally available information where appropriate, and other relevant secondary sources.

Your final output should present an evidence-informed position. This means that, by the end of the research process, you should be able to explain what you have found, evaluate the strength and relevance of the evidence, and present a clear argument, conclusion, or answer to your research question.

You should define a clear research focus or research question that is relevant to your apprenticeship pathway, professional context, or final project. Your topic should link to at least one Knowledge, Skill, or Behaviour from your apprenticeship standard, which you should state as part of your Research Process documentation or as part of your Research Output.

Research Process

You must provide clear evidence of how you conducted your research and how your final output developed.

Your process evidence should:

- **show the steps you took to identify, select, evaluate, and use sources.** This may include search terms, databases or platforms used, AI tools (see “Generative AI Guidance” section below), screenshots of search results, search logs, or other evidence that demonstrates how your research developed over time. You may wish to represent this visually if appropriate, for example through a flowchart or PRISMA-style diagram.
- **include a table of your final selected sources**, along with a sample of sources which were rejected, distinguishing peer-reviewed sources, along with justifications for inclusion or exclusion (consider the CRAAP test!)
- **include at least 5 annotated pages from one of your selected peer-reviewed academic papers** to demonstrate how you read and engaged with a key source of your choice. This may be carried out using an appropriate digital tool (e.g. Zotero annotation tools) or by hand using physical printouts which can be photographed for inclusion in your evidence. Highlighting or underlining, along with comments and questions would form part of appropriate annotation evidence. You can demonstrate how your annotations built up over time by using different colour pens (if by hand) or timestamps (if digital).

Screenshots should support specific claims about your process. For example, if you state that a search initially returned 1,057 papers, you should include a screenshot showing that result. If you refined your search terms, excluded sources, or changed your research focus, your documentation should briefly explain why.

You are encouraged to discuss your research with your module tutor in order to gain formative feedback and advice as your research evolves, which in turn can feed into your process evidence.

You are also encouraged to collaborate informally with your peers and form “research clusters” if you are researching similar topics. Sharing resources

Submission Requirements

Your document submission(s) **must** be text-based (.docx or .pdf) and not password protected. Any submissions which circumvent the Turnitin plagiarism checks will be marked as a fail (0).

When linking to external resources, ensure they are publicly available. Resources that cannot be accessed will be ignored.

There is no word count limit for this assessment, but rather you should focus on engaging with the appropriate number of sources and producing a well-articulated research output which draws on all of them meaningfully and coherently. For example, a paper which is less than 1000 words may risk being too shallow to sufficiently demonstrate meaningful engagement with your chosen sources, whereas a paper over 5000 words might be going significantly beyond what is required or it may be overly verbose (waffling and repetitive). Focus on quality over quantity in your research output and remember you can show off your research process alongside it.

You should review the “Assessment Criteria” at the end of this document to understand how your work will be marked.

Generative AI guidance

AMBER (permitted)

Use of Generative AI (e.g., MS Copilot or ChatGPT) is **permitted** for this assessment, but not required. If using spelling or grammar assistive tools, be cautious if using any Generative AI features which rewrite non-trivial portions of your work in case your original meaning is lost or your writing ends up being incoherent. You remain fully responsible for the work you submit.

Avoid relying on Generative AI to format your references as these tools can often produce references which do not exist which will significantly undermine the credibility of your work and may be considered academic misconduct. Dedicated reference management tools such as EndNote, Zotero or Mendeley can provide more reliable support for this task.

The following are examples of potential uses of GenAI that would be considered **appropriate** for this assessment:

✓ Developing or refining a research question or topic	Using AI to explore possible research directions or refine a research question. Your responsibility: You remain responsible for determining the final focus, framing, and direction of your work. <u>Requires disclosure as part of the research process evidence requirement of this assessment.</u>
✓ Finding keywords, search terms or search strategies	Using AI tools to identify keywords, search terms, or search strategies to support database searching and improve search effectiveness. Your responsibility: You remain responsible for reviewing and refining search strategies to ensure they are appropriate for your topic, discipline, and assessment. <u>Requires disclosure as part of the research process evidence requirement of this assessment.</u>
✓ Identifying, locating or organising sources	Using AI to suggest potential sources or help organise material you have already found. Your responsibility: You remain responsible for ensuring that all sources used are appropriate, credible, and verified. <u>Requires disclosure as part of the research process evidence requirement of this assessment.</u>
✓ Creating diagrams, visuals or infographics	Using AI to generate or refine visual representations based on your own content, analysis, or interpretation. Your responsibility: You remain responsible for the accuracy, meaning, and interpretation of any visual material submitted and should not present AI-generated data, results, or analysis as your own i.e. declare what tools you used.
✓ Agentic AI coding	Using Agentic AI to develop a bespoke website to present your research. Your responsibility: You remain responsible for ensuring the website does not have security vulnerabilities and is accessible for users on different types of devices.
✓ Checking and/or improving	Using AI as a proofreading or editing aid to identify spelling, grammar, punctuation, or typographical errors.

spelling, grammar and punctuation	Your responsibility: You remain responsible for ensuring that any suggested changes accurately reflect your intended meaning and do not introduce new content or ideas.
✓ Improving clarity and readability	Using AI to suggest clearer phrasing, improve expression, or enhance the readability of text. Your responsibility: You remain responsible for the meaning, argument, and interpretation presented in your work.

If you are in doubt about the appropriateness of using any specific assistive technology, please discuss with your module organiser.

For more guidance, please see <https://www.qmul.ac.uk/library/academic-skills/student-guide-to-generative-ai/>.

Late Submissions and Extenuating Circumstances

Assessments submitted after the published deadline will be marked as late and will receive a penalty, unless you have approved Extenuating Circumstances (EC).

You may submit up to seven calendar days after the deadline but a penalty of 5% of the total marks available will be applied to the assessment for each 24-hour period (or part of it) that the work is late (e.g. five marks deducted each day from a total of 100). After seven calendar days the assessment will be recorded as a non-submission and given a mark of zero.

Late penalties are applied automatically so please be aware that even submitting 10 seconds late (e.g. 23:59:10) will trigger a late penalty.

For more information, you can find the latest Assessment Handbook and Extenuating Circumstances policy via the QMUL policies page here: <https://www.qmul.ac.uk/governance-and-legal-services/policy/policies-by-category/>

Knowledge, Skills, and Behaviours (Apprenticeship v1.2)

The nature of this assessment means that many different KSBs may be covered depending on your apprenticeship specialism and choice of research topic. Nonetheless, the following KSBs are closely aligned:

- S13 – Report effectively to colleagues and stakeholders using the appropriate language and style, to meet the needs of the audience concerned.
- S14 – Research, investigate, and evaluate innovative technologies or approaches in the development of a digital and technology solution.
- B4 – Commits to continuous professional development; maintaining their knowledge and skills in relation to developments in digital and technology solutions that influence their work.
- B7 – Maintains awareness of trends and innovations in the subject area, utilising a range of academic literature, online sources, community interaction, conference attendance and other methods which can deliver business value.

Assessment Criteria

Criteria	0-29 (F) Not successful	30-39 (E) Below required standard	40-49 (D) Pass	50-59 (C) Satisfactory	60-69 (B) Good	70-79 (A) Excellent	80-100 (A+) Outstanding
Minimum pass requirement	Both documents must be submitted in text format, such as .docx or .pdf, so that Turnitin plagiarism checks are not circumvented. Submitting the research output but not the research process evidence (or vice versa) will result in a fail (0). All assessment criteria must be passed to be able to pass the overall assessment.						
Research Framing and Direction (20%)	Not attempted or incoherent.	Research focus unclear, too broad, poorly scoped, or weakly connected to the professional context.	A basic research focus or question is provided. Scope, purpose, audience, or professional relevance may be limited or only partially developed.	Research focus is mostly clear, suitably scoped, and relevant to the apprenticeship pathway, professional context, or final project area. The output has a generally clear purpose and direction.	Research focus is clear, well scoped, and professionally relevant. The research question, purpose, and direction provide a strong foundation for the final output.	In addition to B-grade criteria: The framing shows strong judgement in defining a focused, worthwhile, and professionally meaningful area of enquiry, with clear links to apprenticeship context, KSBs, or final project.	In addition to A-grade criteria: Demonstrates mature awareness of the research's professional, technical, ethical, or organisational significance.
Use and Evaluation of Evidence (20%)	Not attempted, incoherent, or wholly reliant on low-quality sources.	Some sources are used, but they are narrow, weakly selected, poorly evaluated, or insufficiently connected to the research question. Academic or professional evidence may be largely absent.	A basic range of relevant sources is used, including some credible academic, professional, technical, regulatory, or industry material. Evaluation is limited or mainly descriptive.	Evidence base is appropriate and generally credible, with reasonable breadth and relevance. Evaluation gives some attention to credibility, quality, or limitations of sources.	Evidence base meets the specified minimum quantitative and qualitative requirements in the assessment brief. Source quality, credibility, relevance, and limitations are evaluated clearly and used effectively.	In addition to B-grade criteria: Evidence is selected and evaluated with strong critical judgement which considers the wider context of the research.	In addition to A-grade criteria: Goes well beyond what was taught in source selection, evaluation, and application. Quality of research beyond what is expected at undergraduate level.

Assessment Criteria

Criteria	0-29 (F) Not successful	30-39 (E) Below required standard	40-49 (D) Pass	50-59 (C) Satisfactory	60-69 (B) Good	70-79 (A) Excellent	80-100 (A+) Outstanding
Critical Analysis and Argument (20%)	Not attempted or incoherent.	Analysis is very limited or mostly descriptive. Links between evidence, claims, and conclusions are weak, unclear, or unsupported.	Some analysis is attempted and a basic position or conclusion is present. Argument, comparison of perspectives, or consideration of limitations may be missing or underdeveloped.	Analysis is generally clear and relevant. Evidence is interpreted to support a developing argument, with some comparison of perspectives and recognition of limitations or uncertainty.	Analysis is clear, coherent, and well supported. The output compares perspectives, evaluates implications, recognises limitations, and develops a persuasive evidence-informed argument or conclusion.	In addition to B-grade criteria: Strong argument, critically informed. Thoughtful consideration of competing interpretations, trade-offs, implications, or constraints.	In addition to A-grade criteria: The analysis demonstrates exceptional insight, originality, or professional judgement.
Research Process Evidence (20%)	Not attempted, lacking transparency, or incoherent. Does not provide any reading and annotation evidence.	Some process evidence is present but very limited, fragmented, or difficult to follow.	Basic process documentation is provided, including some evidence of search activity, source selection, annotation, or tool use. Decisions and links between the research and the final output may be missing or unclear.	Process documentation is mostly clear and includes some relevant evidence such as search terms, search logs, annotated sources, screenshots, diagrams, drafts, or notes on source selection. Some decisions and links between the research and the final output are evident.	Process documentation is clear, comprehensive, and transparent. Shows how sources were identified, selected, evaluated, and used. Meaningful annotation evidence included, in line with the minimum requirements (5 pages) Decisions and links between the research and the final output are clear.	In addition to B-grade criteria: The process evidence provides a strong audit trail of decision making, including refinement of the research process, inclusion and exclusion decisions, source evaluation, annotated academic material, and responsible checking of any tool-assisted work. May go well beyond the minimum requirements of annotation evidence demonstrating strong engagement with source text(s).	In addition to A-grade criteria: Research evidence demonstrates a capability well beyond what is expected at undergraduate level. Research is potentially of publishable quality. Innovative use of appropriate research tools, fully justified.

Assessment Criteria

Criteria	0-29 (F) Not successful	30-39 (E) Below required standard	40-49 (D) Pass	50-59 (C) Satisfactory	60-69 (B) Good	70-79 (A) Excellent	80-100 (A+) Outstanding
Communication and Professional Presentation (20%)	Submitted documents are incomplete, inaccessible, or wholly inappropriate to a professional context.	Submitted documents are difficult to follow, lack structure, or have other substantial issues which impede understanding.	The content of the submitted artefacts is understandable. Structure, referencing, accessibility, or professional presentation may be inconsistent.	Communication throughout the documentation / artefacts is generally clear and well structured. An appropriate paper format or online tool is chosen and used satisfactorily but with mistakes or omissions. Referencing of sources mostly adheres to the chosen format. Overall, submission requires significant refinement.	Communication is clear, well structured, and professional. An appropriate paper format or online tool is chosen and used effectively, with minimal mistakes or omissions. Referencing of sources adheres consistently to the chosen format.	In addition to B-grade criteria: The output is highly polished and engaging, using the chosen format effectively to communicate complex ideas with clarity, coherence, and professional credibility.	In addition to A-grade criteria: Use of paper format or online tool demonstrates an expert ability to communicate ideas effectively.